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1 growth rates

we learned about algorithms, however, we don't really know how to compare them – which one is more efficiency, or fast, etc. in this section we introduce the big-o notation, which is used to measure the growth rate of an algorithm.

1.1 big-o notation

roughly speaking, the big-o notation describes how fast a function $f(x)$ grows – and the growth of this function will never exceed the growth of another function $g(x)$.

1.1.1 definition

let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. we say that $f(x)$ is $O(g(x))$ if there are constants C and k such that $|f(x)| \leq C|g(x)|$ whenever $x > k$. we read this as: "$f(x)$ is big-o of $g(x)$".

in simpler words, the definition of the growth rate of $f(x)$ is $O(g(x))$ means that $f(x)$ grows slower than some fixed multiple(C) of $g(x)$ as x grows without bound past k.

we call the constants C and k the witnesses – and there are infinite witnesses that exist when $f(x)$ is $O(g(x))$. if we find a pair of witnesses (C, k) that work, then any C' such that $C' > C$ will also work; when we use the same k. likewise, any k' such that $k' > k$ will work, when using the same C.

1.1.2 finding the witnesses

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1.2 big-omega notation

the big-o notation has its own limitations, however. it only tells us how fast the function grows – no faster than $g(x)$ (an upper bound). sometimes, we want to know if the function grows at least as fast as $g(x)$ (a lower bound). in computer science, we use big-o as a measurement of the "worst case", and big-omega as a measurement of "best case".

1.2.1 definition

let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. we say that $f(x)$ is $\Omega(g(x))$ if there are constants C and k with C *positive* such that $f(x) \geq C|g(x)|$ whenever $x > k$. we read this as: "$f(x)$ is big-omega of $g(x)$".

aside: there is a connection between big-o and big-omega. specifically, $f(x)$ is $\Omega(g(x))$ *if and only if* $g(x)$ is $O(f(x))$.

1.3 big-theta notation

1.3.1 definition

let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. we say that $f(x)$ is $\Theta(g(x))$ if $f(x)$ is $O(g(x))$ and $f(x)$ is $\Omega(g(x))$. when $f(x)$ is $\Theta(g(x))$, we say that f is big-theta of $g(x)$, that $f(x)$ is of *order* $g(x)$, and that $f(x)$ and $g(x)$ are of the *same order*.